THE EXACT SEQUENCE RELATING THE CONORMAL SHEAF OF A SURJECTION OF SHEAVES OF ALGEBRAS TO THE MODULES OF DIFFERENTIALS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The functoriality of the module of differentials for a commutative diagram of sheaves of rings, with the map characterised on differentials of local sections. Author lemma lemma-functoriality-differentials, whose statement is reproduced verbatim as a context span of this card (source lines 5136-5157). Supplies the situation in which the statement is formulated and the right hand map of the sequence.
- The Leibniz rule for the universal derivation, and the compatibility of the formation of differentials with taking stalks. Used inside the counted proof (source lines 5184-5191 and 5195-5197). Shows that the displayed rule defines a map on the quotient by the square of the kernel and identifies the sequence with the algebraic one.

Source reference. Original lemma lemma-abelian: sources/stacks-modules/pinned-original-source.tex, source lines 439–439. Navigation pointer only: the counted text refers to the Stacks item lemma-abelian, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-stalk-module-differentials: sources/stacks-modules/pinned-original-source.tex, source lines 5197–5197. Navigation pointer only: the counted text refers to the Stacks item lemma-stalk-module-differentials, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged

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Source reference. Original reference algebra-lemma-differential-seq: sources/stacks-modules/pinned-original-source.tex, source lines 5199–5199. Navigation pointer only: the counted text refers to the Stacks item algebra-lemma-differential-seq, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *In Lemma 0.2 suppose that $\mathcal{O}_2 \to \mathcal{O}'_2$ is surjective with kernel $\mathcal{I} \subset \mathcal{O}_2$ and assume that $\mathcal{O}_1 = \mathcal{O}'_1$. Then there is a canonical exact sequence of $\mathcal{O}'_2$-modules*

$$\mathcal{I}/\mathcal{I}^2 \longrightarrow \Omega_{\mathcal{O}_2/\mathcal{O}_1} \otimes_{\mathcal{O}_2} \mathcal{O}'_2 \longrightarrow \Omega_{\mathcal{O}'_2/\mathcal{O}_1} \longrightarrow 0$$

The leftmost map is characterized by the rule that a local section f of $\mathcal{I}$ maps to $df \otimes 1$.

Lemma 0.2. *Let X be a topological space. Let*

$$\begin{array}{ccc} \mathcal{O}_2 & \xrightarrow{\varphi} & \mathcal{O}'_2 \\ \uparrow & & \uparrow \\ \mathcal{O}_1 & \longrightarrow & \mathcal{O}'_1 \end{array}$$

be a commutative diagram of sheaves of rings on X. The map $\mathcal{O}_2 \to \mathcal{O}'_2$ composed with the map $d : \mathcal{O}'_2 \to \Omega_{\mathcal{O}'_2/\mathcal{O}'_1}$ is a $\mathcal{O}_1$-derivation. Hence we obtain a canonical map of $\mathcal{O}_2$-modules $\Omega_{\mathcal{O}_2/\mathcal{O}_1} \to \Omega_{\mathcal{O}'_2/\mathcal{O}'_1}$. It is uniquely characterized by the property that $d(f) \mapsto d(\varphi(f))$ for any local section f of $\mathcal{O}_2$. In this way $\Omega_{-/-}$ becomes a functor on the category of arrows of sheaves of rings.

Proof. For a local section f of $\mathcal{I}$ denote $\overline{f}$ the image of f in $\mathcal{I}/\mathcal{I}^2$. To show that the map $\overline{f} \mapsto \mathrm{d}f \otimes 1$ is well defined we just have to check that $\mathrm{d}f_1 f_2 \otimes 1 = 0$ if f_1, f_2 are local sections of $\mathcal{I}$. And this is clear from the Leibniz rule $\mathrm{d}f_1 f_2 \otimes 1 = (f_1 \mathrm{d}f_2 + f_2 \mathrm{d}f_1) \otimes 1 = \mathrm{d}f_2 \otimes f_1 + \mathrm{d}f_1 \otimes f_2 = 0$. A similar computation show this map is $\mathcal{O}'_2 = \mathcal{O}_2/\mathcal{I}$-linear. The map on the right is the one from Lemma 0.2. To see that the sequence is exact, we can check on stalks (Lemma lemma-abelian). By Lemma lemma-stalk-module-differentials this follows from Algebra, Lemma algebra-lemma-differential-seq. □